Development of a regression model to forecast apartment rent prices

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***Abstract*—This project presents the development of a regression model to forecast monthly apartment rent prices using a dataset of nearly 100,000 listings across the United States. The approach involved extensive preprocessing, feature engineering from textual data, and outlier handling. Several tree-based regression models were evaluated, with Extra Trees and Random Forest achieving the best performance in terms of Mean Absolute Error. The proposed methodology yielded promising results, highlighting the potential of feature engineering from textual data, while still leaving room for future enhancements.**

I. Problem Overview

The goal of this project is to develop a regression model that accurately predicts the monthly rent price of an apartment based on various features. The dataset used comprises 99,487 records collected from various sources across the United States, each representing a unique apartment. The dataset is divided into:

- development set, containing 79,589 records and their values for 21 features, one of them being the *price* target variable
- evaluation set, containing 19,898 records and the same features of the development set, minus the target variable

Both datasets are provided in .csv format. The development dataset was employed to build a regression model to predict the *price* feature, which was then used to make predictions on the evaluation dataset. The performance of the model was assessed in terms of the Mean Absolute Error (MAE):

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

where:
y_i: actual value of the i-th observation
$\hat{y}_i$: predicted value of the i-th observation
n: number of observations

In order to choose the appropriate preprocessing procedure, a preliminary data exploration step was carried out. Figure 1 shows the number of apartment listings by state in the development dataset. Focusing on categorical and numerical features, by visually comparing the data distribution for each feature of the two datasets, they were assessed to be similar, as shown in figures 2 and 3. The inspection of textual data also provided useful insight on how to estimate missing features, as described in section II-A. The dataset posed several challenges due to its heterogeneity and the presence of missing or noisy data. In particular, the unstructured nature of textual fields such as *title* and *body* required careful preprocessing and feature engineering to ensure model robustness.

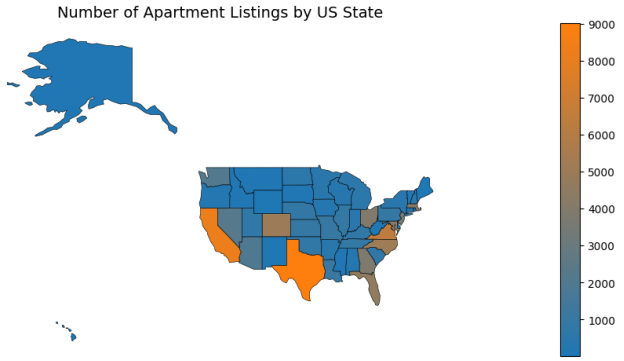


Fig. 1. Number of apartment listings by US state in the development dataset

II. Proposed Approach

A. Preprocessing

Following the initial observations, the proposed approach aims at detecting and eliminating outliers and features that do not contain useful information, handling missing values, and transforming textual data into a more appropriate format for regression.

a) Data reduction. In evaluating the usefulness of each feature for the regression task, it was determined that the *address* feature could be discarded, as it contains a substantial number of missing values: 73,226 out of 79,589 in the development dataset, and 18,320 out of 19,898 in the evaluation dataset. The *price_type* and *currency* features were also removed, since they hold only a single value across both datasets and therefore do not contribute meaningful information to the model. Regarding the *cityname* and *state* features, it was noted that they encode information already captured by the *latitude* and *longitude* features. Although imputing their missing values is technically feasible, it introduces additional challenges. Specifically, the *cityname* feature presents a very large domain, making it difficult to encode without incurring dimensionality issues. In contrast, the state feature has a more limited domain, but experiments showed that including it did not improve model performance. However, a possible future

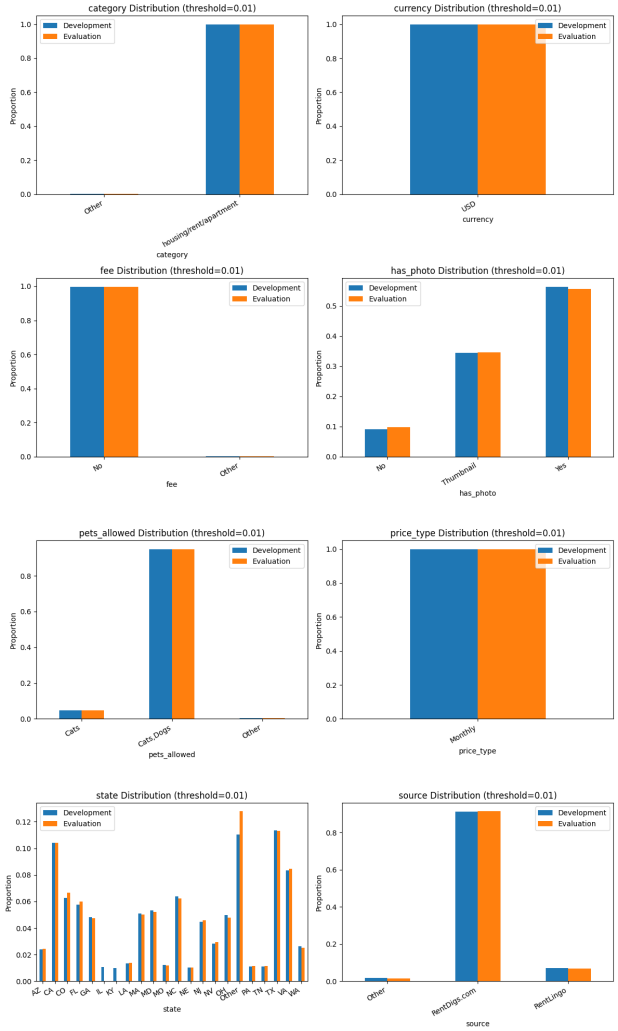


Fig. 2. Comparative histograms of normalized frequency distribution of categorical features

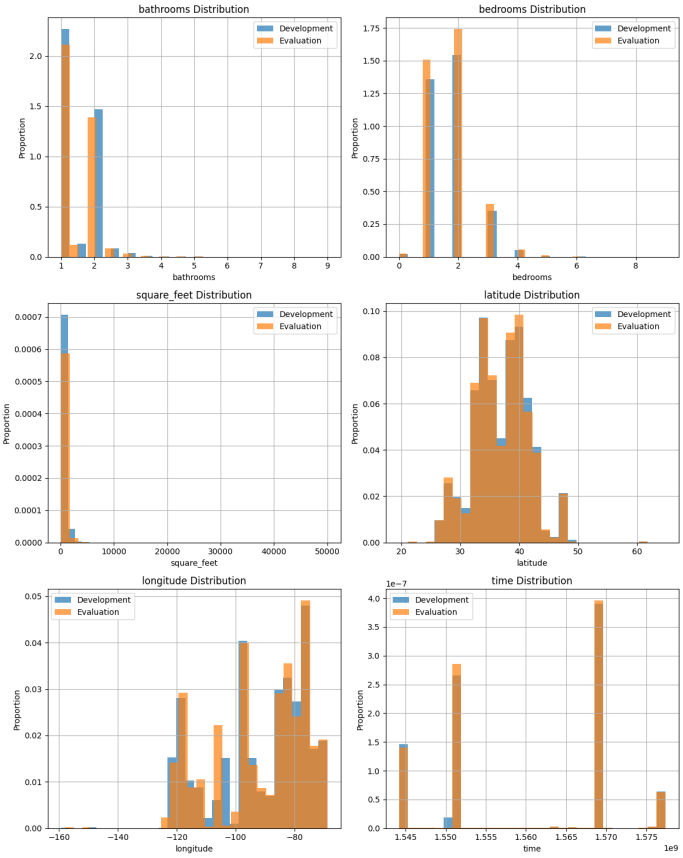


Fig. 3. Comparative histograms of normalized frequency distribution of numerical features

approach could involve estimating the missing values of *cityname* and *state* from the available *latitude* and *longitude* coordinates, and then applying clustering techniques to *cityname*, as suggested in [1], and using one-hot encoding for the inferred *state*. Similarly, the features *category*, *has_photo*, *pets_allowed*, *source*, and *fee* were also discarded, as empirical evaluations indicated that their inclusion did not lead to any meaningful performance improvements.

b) Handling missing values. Several features of both datasets contained missing values, which were handled differently.

- *amenities* and *pets_allowed*: missing values were imputed with the strings "none" and "no", respectively. This choice is grounded in the assumption that the absence of data in these fields is not due to a data acquisition error, but rather reflects the non-existence of the attribute. For instance, landlords typically highlight the presence of desirable amenities or pet-friendly policies to attract tenants; conversely, if these fields are empty, it is more plausible that no amenities are offered or that pets are not allowed. This approach avoids introducing artificial noise that might arise from applying generic imputation techniques such as one-hot encoding with missing as a separate category.
- *bathroom* and *bedroom*: missing values for these features were imputed based on the corresponding *square_feet* values, leveraging their strong correlation as also suggested in [2]. The underlying assumption is that the number of bathrooms and bedrooms is strongly correlated with the size of the apartment, as supported by the correlation matrix shown in Figure 4. To leverage this relationship, apartments in the development dataset were divided into ten equally populated bins based on square footage quantiles. For each bin, the average number of bathrooms and bedrooms was computed using available data. These averages, rounded to the nearest integer, were then used to impute missing values for both features in both the development and evaluation datasets.
- *latitude* and *longitude*: different strategies were adopted depending on the subset of the dataset. It is worth noting that instances missing the *latitude* and *longitude* values, were also lacking values for *cityname* and *state*, making a reliable imputation of geographic coordinates

particularly challenging. To address this issue, a function was developed to extract location-related information, such as ZIP codes, city names, and state abbreviations, from the *title* and *body* fields. This extracted information was then used to perform forward geocoding via the Nominatim sofware. The function was initially tested on both the development and evaluation datasets. However, its application to the development set led to a degradation in performance. This deterioration is likely due to the introduction of noise caused by imprecise geocoding results. For this reason, in the final version of the project, the function was applied only to the evaluation dataset, where it showed modest improvements, while records in the development set with missing *latitude* or *longitude* were discarded to preserve data quality.

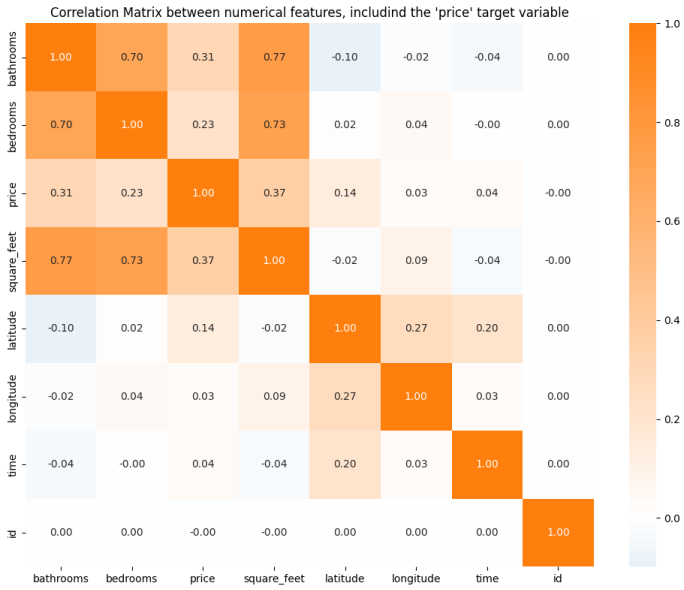


Fig. 4. Spearman correlation matrix between numerical features

c) Handling textual data.

- *amenities.* This feature consists of a comma separated list of facilities available in each apartment. To make this information suitable for regression models, each record was first transformed into a list of individual amenities. Missing values were treated as empty lists. A multi-label binarization technique was then applied, converting each list into a binary vector that captures the presence or absence of each amenity. Importantly, the development and evaluation datasets share the same set of possible amenities, ensuring consistent feature representation across both. This transformation highlights the value of extracting structured signals from semi-structured fields in real estate data—a strategy that has also proven effective in prior research [3].
- *title* and *body.* By inspecting a sample of the *title* field, we noticed the presence of dollar amounts in approximately 10% of the development set. In about 90% of these cases, the extracted value matched the ground truth. The body field, on the other hand, typically contained a rent range, but it also included other numerical values (e.g., heating costs). To better capture this information, we engineered four new features from both the *title* and *body* fields: minimum, maximum, mean, and standard deviation, computed using only the numeric values preceded by a dollar sign. These values were extracted using regular expressions. In cases where no such values were found, the corresponding features were set to zero. The inclusion of numerical features extracted from the *title* and *body* fields significantly improved model performance, confirming that, as noted in [4], numerical expressions in textual descriptions carry valuable semantic weight in property valuation tasks.

d) Outliers detection and removal. As shown in Figure 5, two observations in the development dataset appear to be potential outliers. The first has a price of 50,000, while the second exhibits an unusually large square footage of 40,000. We decided to remove the former due to its isolated nature in the price distribution, whereas the latter was retained, since the evaluation dataset includes an instance with a square footage as high as 500,000.

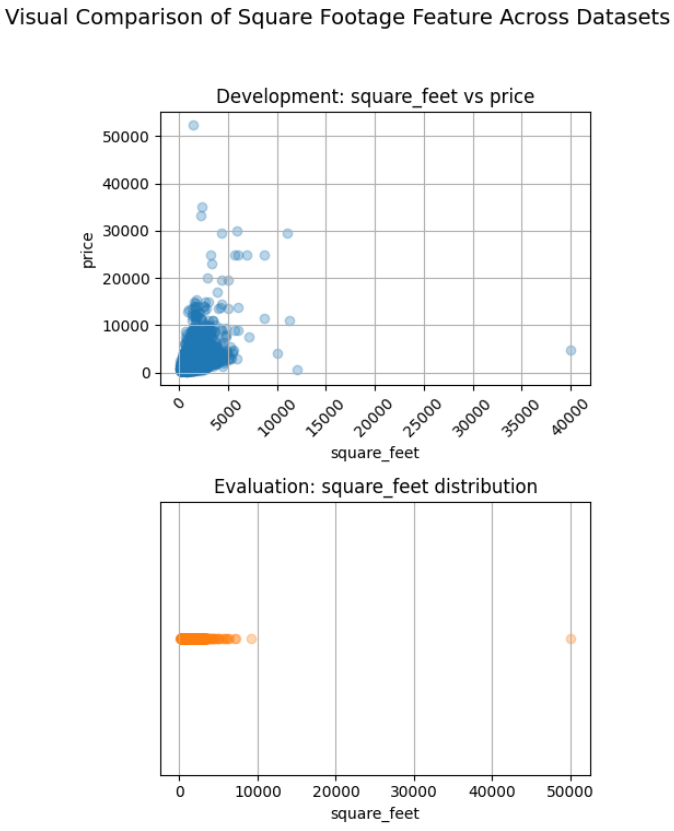


Fig. 5. Comparison of square footage distributions in development and evaluation datasets for outlier detection purposes

B. Model selection

We initially tested several models on the development dataset using their default hyperparameters to obtain a preliminary assessment of their effectiveness for the task at hand. The dataset was divided into two subsets: 80% for training and 20% for testing. The models considered included Random Forest, Extra Trees, XGBoost, HistGradientBoosting, CatBoost and Gradient Boosting. These tree-based models are well-suited to handle non-linear relationships and heterogeneous features. Feature standardization did not lead to any performance improvements; therefore, it was not applied. Models performances were evaluated using the Mean Absolute Error (MAE). Results for this preliminary step are shown in Table I.

TABLE I
MODELS' PERFORMANCES WITH DEFAULT HYPERPARAMETERS

Model	MAE
Extra Tree	152.488
Random Forest	154.746
XGBoost	191.756
Hist Gradient Boost	214.267
Catboost	238.189
Gradient Boost	278.300
Baseline	339.641

C. Hyperparameters tuning

After evaluating the initial performance of the different models, we performed hyperparameter tuning for each of them using a Grid Search approach, using the same slit percentage mentioned in II-B. The range of tested hyperparameter values, along with the selected optimal configurations for each model, are reported in Table II.

TABLE II
HYPERPARAMETERS TUNING

Model	Parameters	Values	Selected
Random Forest	max depth	[None, 10, 20]	None
Extra Tree	max depth	[None, 10, 20]	None
XGBoost	max depth learning rate gamma	[3, 5] [0.05, 0.1] [0, 0.1]	5 0.1 0
Hist Gradient Boost	learning rate max leaf nodes	[0.05, 0.1] [31, 63]	0.1 63
Catboost	depth learning rate l2 leaf reg	[4, 6] [0.05, 0.1] [3, 5]	6 0.1 3
Gradient Boost	learning rate max depth min samples leaf	[0.05, 0.1] [3, 5] [1, 2]	0.1 5 1

III. RESULTS

As reported in Table III, the Extra Trees Regressor outperformed all other models in terms of predictive accuracy. Consequently, a final Extra Trees Regressor, configured with the optimal hyperparameters, was trained on the entire development dataset and used to predict the target *price* variable on the evaluation dataset, achieving a MAE score of 136.267. The corresponding feature importance scores are shown in Table IV. The most impactful features include the geographical coordinates, apartment size, and textual attributes derived from the *body* and *title* fields, highlighting the crucial role of textual descriptions in accurately estimating rental prices.

TABLE III
MODELS' PERFORMANCES WITH SELECTED HYPERPARAMETERS

Model	MAE
Extra Tree	142.692
Random Forest	148.095
Hist Gradient Boost	196.229
Gradient Boost	201.419
XGBoost	203.869
Catboost	232.624
Baseline	339.641

TABLE IV
FEATURE IMPORTANCE SCORES COMPUTED USING THE EXTRA TREES REGRESSOR

Feature	Percentage importance
longitude	23.91
latitude	11.98
square_feet	11.37
bathrooms	10.70
time	5.40
min_from_body	4.03
mean_from_body	3.42
max_from_body	2.51
bedrooms	2.42
id	1.70
Gym	1.38
std_from_body	1.25
Elevator	1.25
Pool	1.25
Parking	1.17
mean_from_title	1.07
max_from_title	1.02
min_from_title	1.01
Playground	0.98
TV	0.97

IV. DISCUSSION

As shown in Table III, our model consistently outperforms the established baseline, benefiting from both the effective feature extraction strategy described in Section II-A and the strong predictive capabilities of the selected Extra Trees Regressor. The feature importance analysis highlights the central role of geographic coordinates and apartment size in determining rent prices. Interestingly, several features derived from textual descriptions also ranked highly, confirming that unstructured data contributes significantly to model performance. Future improvements could include integrating pre-trained language models to better capture semantic patterns in textual data, or applying clustering techniques to group listings by urban area, thus enriching the spatial representation beyond raw coordinates, as described in [5].

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